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Sky Track-Aviation Data and Pipeline Analysis

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Sky Track-Aviation Data and Pipeline Analysis

By

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GRADUATECAPSTONE SEMINAR PROJECT

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ABSTRACT

The aviation industry generates a significant amount of data from various sources, such as flight operations, passenger information, and logistics. Effectively analyzing this data is crucial for improving operational efficiency, making accurate forecasts, and supporting data-driven decision-making. This project focuses on designing and implementing an end-to-end data engineering pipeline using Azure cloud services, enabling efficient data processing, transformation, and analysis, with insights delivered through Power BI. The solution aims to address the growing need for aviation data analysis by creating a scalable and automated ETL pipeline that can integrate data from multiple sources. The motivation behind this project stems from the challenges the aviation industry faces in managing and analyzing large-scale data that is often siloed across different systems. Traditional data management infrastructures struggle to handle the vast volume, variety, and velocity of data generated in the aviation sector. These limitations can prevent timely access to actionable insights, affecting operational efficiency and safety. By creating a robust ETL pipeline, this project seeks to break down these data silos, enabling seamless data integration, transformation, and processing to improve decision-making.The proposed solution will leverage Azure Data Factory for extracting and transforming data from various aviation-related sources, Azure Data Lake for scalable storage, and Power BI for real-time analytics and visualization. The data will be structured using a star schema to optimize query performance, allowing for the analysis of key aviation metrics such as flight delays, fuel consumption, and passenger trends. The project will also implement a CI/CD pipeline using Azure DevOps, ensuring that updates and deployments are smooth and continuous. Ultimately, this solution will provide a comprehensive approach to aviation data analysis, enabling real-time insights that can enhance operational efficiency, reduce delays, and improve overall passenger experience.

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1 Project Description

The project involves designing and deploying a modular, data driven, and automated data building and analysis system that will mine and process aviation operational data in a bid to increase efficiency, reduce delays, decrease fuel consumption and better management of fleet as a whole. This enhances operational efficiency within the aviation businesses as they will be looking for insight based on the facts provided by the data generated.

However, these companies cannot remain stagnant because of the intense competition in the market as far as cost and operational efficiency is brandier. In the course of operations, enormous amounts of data are produced from various operational systems such as flight operations, maintenance logs, fuel management systems, and crew scheduling. Nevertheless, such databases are incapable of addressing the multi-dimensional, high volume and time sensitive nature of such data: resulting to a disjointed system of information that cannot be easily retrieved or used in any meaningful way. This is just what this project is going to do, build in place an effective and holistic data pipeline for combining such operational data as would typically be hard to procure and making use of the combined data in the optimization of various managerial processes.

Solution:

Building on the concept of this project, this project will utilize Microsoft Azure Cloud Services and implement a complete ETL (Extract, Transform, Load) solution with regard to the aviation operational data. In addition, the solution will enable operational efficiency by providing up to the minute evaluation of important measures complemented by brief communication over the same.

Azure Data Factory (ADF)will help to orchestrate the ETL activities from various Api sources, such as:

- Flight scheduling and operations systems
- Maintenance records and sensor data from aircraft
- Fuel consumption and tracking systems
- Weather data and its impact on flight performance
- Crew scheduling and duty records

Azure Data Lake Storage (ADLS) will provide storagewarehouse for both structured and unstructured data, ensuring that the system can handle large volumes of aviation data.

Azure Data bricks will be used for processing and transforming data into a star schema for optimized query performance, allowing for efficient analysis of key operational metrics.

Power BI will be used to provide real-time dashboards for tracking operational KPIs such as:

- **Flight Delays:** Analysis of causes, frequency, and patterns to enable productive scheduling and minimize delays.
- **Fuel Efficiency:** Real-time monitoring of fuel consumption per route, aircraft type, and environmental factors to optimize fuel usage and carbon emissions.

Aircraft Utilization: Tracking the efficiency and uptime of aircraft, ensuring optimal utilization and reducing maintenance downtime.

Crew Efficiency: Analyzing crew scheduling and duty hours to enhance efficiency and comply with regulations.

Azure DevOps will be used to implement a CI/CD pipeline, ensuring smooth and continuous integration and deployment of updates, data pipelines, and analytics models.

Data Structure and Storage:

- The star schema will be used to structure the data for fast and efficient querying. This will allow the system to support complex, multi-dimensional analysis on operational metrics, ensuring that the aviation companies can extract actionable insights quickly.

Key Features:

- Optimized ETL Pipeline: It should be able to accommodate the large operational data and at any point the most recent data will be available for analysis.

- Real-Time Operational Insights: This way, in an aviation company, instead of making decisions just by a gut feeling, the company will have real time dashboards and reports on how the operation is performing.

- Predictive Analytics: This system can further be expanded to include predictive models of when maintenance is going to be required, how much fuel will be burned up, and how long delays are going to last, all based on historical and real-time data.

- Cost Optimization: This solution will also give information on items like fuel consumption and maintenance intervals so that operational costs can be reduced to a minimum.

- Seamless Data Integration: This pipeline will end all the data silos by pulling in operational data from all different systems into one system that will feed real-time data to all the teams.

Expected Outcome:
By implementing this solution, aviation companies can:

- Fewer flight delays, better use of scheduling and resources.
- Optimize flight paths, aircraft usage, and in turn fuel costs through data.
- Optimize fleet management --analyze aircraft performance and maintenance needs in real time.
- Use crew more efficiently, with better crew scheduling and duty management analysis, within the rules of course, but to a more productive level.

1.1 Competitive Information

IBM Maximo for Aviation

- Description: IBM Maximo is a complete aviation asset management tool that allows airlines to keep tabs on and maximize maintenance schedules, parts inventory, and fleet performance.

- Features: Optimizing maintenance, asset lifecycle management, aviation regulation compliance, IoT and AI predictive maintenance.

- Competitive Advantage: IBM Maximo is a very mature product with a good history in asset management as well as in predictive analytics. But that might be focused more on asset management, and not so much overall operational data integration and real-time analysis across marketing, sales, human resources, etc.

GE Aviation's Predix Platform

- Description: Predix is a cloud based company that specializes in predictive maintenance, fuel efficiency optimization and real time flight data analytics. It focuses on maintaining aircrafts and conserving fuel usage, by utilizing IoT sensors and data analytics.

- Features: Preventive maintenance, live in flight tracking, fuel efficiency analysis, AI-derived insights.

- Competitive Advantage: GE's Predix platform is an aviation focused platform with predictive maintenance and IoT integration as a major focus. That's what makes it so difficult, because its so focused on maintenance and all the IoT data that it might be missing the bigger operational picture, crew management, delays, etc.

3. SITA OptiFlight

- Description: SITA Opti Flight is an advanced flight planning tool used to aid in fuel consumption and ultimately the reduction of the impact on the environment. It offers real-time data on flight paths and optimal fuel usage.

- Features: Flight planning, fuel optimization, real-time weather data integration, route optimization.

- Competitive Advantage: SITA is the most specialized of all in flight path and fuel optimization, and will therefore be the greatest competitor in fuel efficiency metrics. It doesn't do quite as much as the general operations of the proposed project, but it are much more specialized.

1.2 Relationship to Other Applications/Projects

- Existing Aviation Projects: This project fits nicely with airline operations management systems and predictive maintenance initiatives in that it combines information from maintenance, crew scheduling, and fuel management to provide a comprehensive look at the entire operation.

- Data Engineering Projects: It is based on ETL processes and real-time analytics methodologies that are employed in other industries but applied to aviation data.

- Cross-Industry Applications: The integration to IoT and data visualization tools (Power BI) allows for better analysis and display of insights.

1.3 Assumptions and Dependencies

·Describe any assumptions made (e.g., new functionality or capability that will provided by other in the same time-frame of your Project).

Other projects, applications, services Functionality and interfaces provided by this project/application/service/capability that this application /service is dependent on

Add any necessary development and/ or customer operational procedural changes required to support this project ·

1.4 Future Enhancements

Discuss any planned or possible evolution of the project. Some project will need to be delivered in phases; this will be the place to document your plan.

1.5 Definitions and Acronyms

Acronym items should be included here. For each special term supply a definition here.

2 Project Technical Description

This project will create a robust, scalable and automated data engineering pipeline for aviation operational data. The design will bring together multiple data sources to provide real time insights to improve operational efficiency, reduce delays and optimize resource usage.

1. Architecture
Cloud Based: The solution will be hosted on Microsoft Azure to use its scalability and services.
Data Flow: ETL (Extract, Transform, Load) process where data is extracted from multiple sources, transformed for analysis and loaded into a data warehouse for querying and reporting.

2. Components
Azure Data Factory (ADF):
Role: Extracts and transforms data from multiple operational sources (e.g. flight schedules, maintenance logs, fuel tracking).
Functionality: Scheduled data pipelines to automate ETL and support both batch and real time data ingestion.

Azure Data Lake Storage (ADLS):
Role: Scalable storage for structured and unstructured data.
Functionality: Single repository for all operational data, easy to access for analytics.

Azure Data bricks:
Data processing and transformation.
Uses Spark to clean and prepare data for analysis, star schema for querying.

Power BI:
Data visualization and reporting.
Real time dashboards and reports on operational metrics, so stakeholders can make data driven decisions.
Azure Data bricks:

Data processing and transformation.

Uses Spark to clean and prepare data for analysis, star schema for querying.

Power BI:

Data visualization and reporting.

Real-time dashboards and reports on operational metrics, so stakeholders can make data driven decisions.

Azure DevOps:

CI/CD Pipeline.

Smooth integration and deployment of updates, development lifecycle.

Data Sources

Flight Operations Data: Schedules, delay info, performance metrics.

Maintenance Records: Aircraft maintenance logs, parts inventory, sensor data.

Fuel Management Data: Fuel consumption per flight and route.

Crew Scheduling Data: Crew assignments and duty hours.

Environmental Data: Weather affecting flight operations.

Data Modeling

Star Schema: Data will be in star schema to enable querying and analysis of KPIs.

Security and Compliance

Data Governance: Data governance to comply with aviation regulations and data privacy.

Access Control: Role based access control (RBAC) to restrict access to sensitive data.

2.1 Application Architecture

The architecture of the system for aviation data analysis has a number of integrated pieces in place, which should function to efficiently gather and process data from a set of varied sources: Aviation Stack API provides the different aviation-related information; the Bureau of Transportation Statistics API provides essential weather-related data quite relevant for the analysis.

Basically, this is done for data ingestion with Azure Data Factory, in short ADF, and more specifically, the ADF Copy Activity. It orchestrates the process of collecting data coming from these APIs and drops the data into a specified raw storage layer, configured as Azure Data Lake Storage Gen2. This acts like a storage layer-almost a catchment area-where the data ingested is retained in its raw state before processing. Indeed, according to the medallion architecture, the further transformation and structuring of the data in successive layers, summarily referred to as the bronze, silver, and gold layers, reflects progressive stages in which the data goes through processing and refinement.

Azure Data bricks was used for data processing, which aided in cleaning, transforming, and enriching raw data. This step was a delicate part of data preprocessing, where quality and usability were developed further. The refined data will be placed back into Azure Data Lake Storage for storage in a structured manner, optimized for efficient querying and retrieval.

Azure Synapse Analytics is a powerful service that facilitates the analytics phase, enabling complex querying and integration of data from disparate sources. These, in turn, will allow the derivation of insight through the advanced analytics capability. Finally, the analyzed data is visualized using Power BI, where interactive dashboards are built to show insights that are easily consumable. This visualization component is a major enabler of stakeholder decisions based on the insights obtained from the data of the whole system.

Below is the schema for the above description, showing how data will flow seamlessly from the various APIs via ADF Copy Activity into ADLS Gen2. Further processing in Databricks, from where it is to be curated in Azure Synapse Analytics for analysis down to a Power BI dashboard for proper visualization of insights.

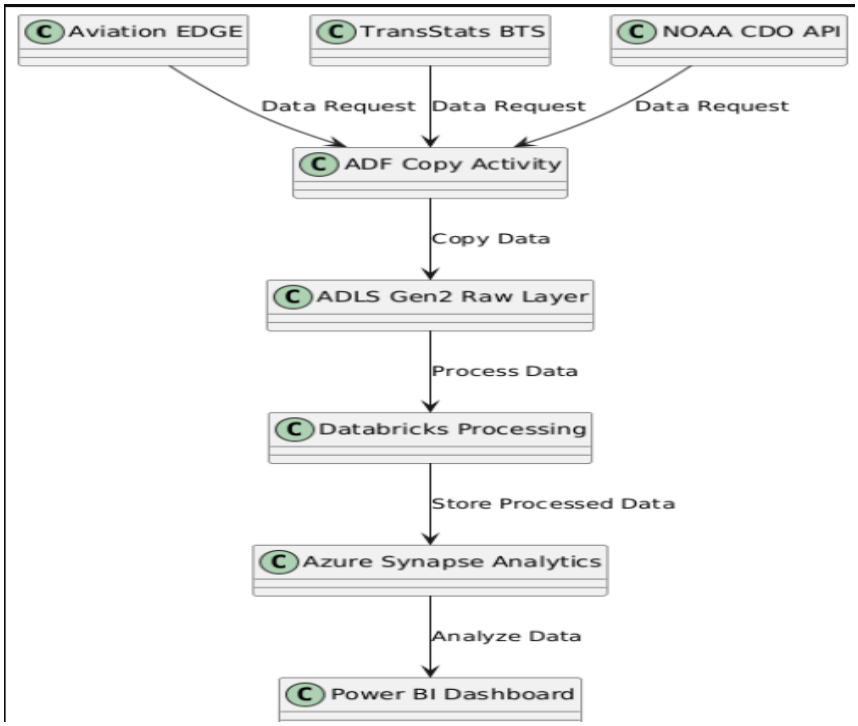


Figure 1Pipeline Architecture

2.2 Application Information flows

First of all, Data bricks Processing processes the raw data in order to have the appropriate quality and format for analysis. It is called the Bronze Layer, indicating the first landing layer for raw data coming from APIs ingested and stored on Azure Data Lake Storage Gen2. This layer is crucial for holding unrefined data that will be transformed into the desired format in further steps.

It is followed by the Silver Layer, where it undergoes cleaning and basic transformations once it has been stored in the Bronze Layer. Data is aggregated in this layer for a more structured format and to support more structured queries and analytics. Finally, the Gold Layer will contain data refined in the Silver Layer with curated data and analytical insights. The focus of this layer would be to build extensive reports summarizing the findings derived from the processed data.

The workflow ends with the Power BI Dashboard, which involves visualizations of data from the Gold Layer. This dashboard should then enable stakeholders to access and interpret the insights derived from the whole processing pipeline for informed decision-making based on analytics derived from the aviation data.

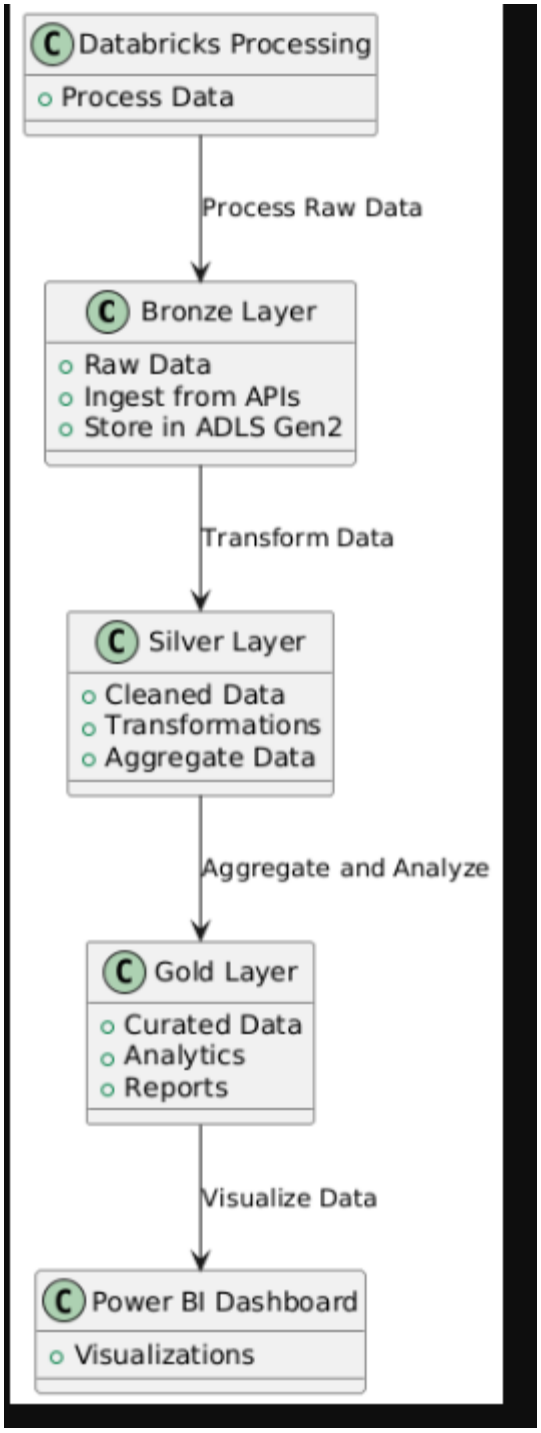


Figure 2DataBricks-Medallion Architecture-Integration with PowerBI

2.3 Interactions with other Projects (if Any)

This is an independent project, Does not have any type of interaction with other Projects directly or indirectly.

2.4 Interactions with other Applications

List all possible interaction with other application, and what are needed to address these interactions.

2.5 Capabilities

- Ingest and integrate data from various APIs that include flight scheduling systems, maintenance logs, fuel management systems, weather, and crew scheduling systems.
- Use batch processing of volume updates as well as real-time streaming of data.
- The ADLS can handle volumes of structured and unstructured data in aviation.
- The storage of data in a well structured format for example, a star schemato make querying and analytics more efficient.
- Data management policies: data quality, consistency, and compliance
- Data integration by Azure Data Factory: Extract data from different sources and transform raw data into useable formats by cleaning, enriching, and transforming. Perform largescale processing and data transformations using Azure Databricks for big and complex data sets efficiently.
- Utilize robust error handling and monitor any errors that might occur during ETL, and log them for auditing.
- Statistical analysis, trending analysis, and predictive modeling are facilitated to gain insight from operational data.
- Power BI integration will be provided to visualize real-time data and present dashboards showing KPIs of operational efficiency.
- This should also provide the functionality to create customized reports based on particular metrics of the operation by different stakeholders.
- Ability to deploy predictive models for maintenance scheduling, fuel consumption forecasting, and delay prediction from historic and real-time data.
- What if analysis supports to find out the consequences of different operational decisions and conditions.
- Role based access controls are to be provided to ensure that sensitive information should only be given to those who require it.
- Data shall be encrypted both during rest and transmission phases to retain its confidentiality and integrity.
- Perform routine audits, monitoring of field compliance, as well as adherence to industry regulations and data governance policies. Utilize available tools that guarantee team collaboration and communication with regards to data insights, findings from analyses, and decisions on operations. Record full documentation of the data pipelines, ETL processes, and analytical methodologies for future reference and onboarding of new analysts.
- Utilize Azure DevOps for CI/CD pipelines that automate deploying updates, data pipelines, and analytics models.
- Apply versioning on data transformation scripts and code to track changes for maintaining integrity within the data. Establish monitoring for pipeline performance, performance in databases, and applications to ensure smooth operations with no issues. Set up alerts on anomalies that would require immediate attention in respect to data processing and analysis. Train the end users to use dashboards and reporting tools to derive analytics effectively.
- Establish a support mechanism through which users can forward system related queries and issues.

2.6 Risk Assessment and Management

Data Quality Risk:

- Poor data quality may lead to incorrect analysis and decision-making.
- Recommendation: Establish appropriate mechanisms for data validation and cleansing in the ETL pipeline to ensure data integrity for analysis.

Integration Challenges:

- Difficulty in integrating data coming from various sources can delay project timelines by a great extent.
- Recommendation: Establish an enterprise API management strategy and conduct comprehensive testing with every data source independently before full integration.

Regulatory and Compliance-related Risks:

- Litigation based on non-compliance with industry regulations is probable.
- Recommendation: Establish data governance policies and regular auditing of data handling practices to ensure compliance with regulations.

Scalability Issue:

- The system will not be able to scale with the volume of data, which might affect performance.
- Recommendation: The architecture should be modular and scalable from the very beginning, based on Azure services that support scaling effortlessly.

User adoption issues:

- Employees' resistance to change may hinder the venture's success.
- Recommendation: There is a need to do extensive training and support for the end user to make them aware of the benefits of the new system that is going to be adopted.

3 Project Requirements

3.1 Identification of Requirements

Data Integration Capability

- The system must integrate data from at least five different operational sources: flight scheduling, maintenance logs, fuel management, weather data, and crew scheduling.

Implementation: Mandatory

- This will ensure a broad-based data pipeline that can support analytics from multiple operational perspectives for better decision-making.

ETL Pipeline Capability

The ETL pipeline has to process and load the data no more than 30 minutes after the generation from operational systems.
Implementation: Mandatory this provides that the data stays timely and relevant for operational metric analysis in near real time.

Data Quality Capability

There should be implementation of data validation checks towards ensuring at least 95% data accuracy before loading into the data lake.
Implementation: Mandatory
This means that for any valid analysis and decision-making, good quality data is required.

Reporting Capability

- The system shall provide real-time dashboard representation of key performance indicators related to flight delays, fuel efficiency, aircraft utilization, and crew efficiency.

Implementation: Required

- This will enable the stakeholders to track operational efficiencies and make correct decisions in the shortest time.

Scalability Capability

The architecture should be able to scale up and support at least a 200% increase in the volume of data for at least the first year of service.
State: Mandatory In this way, it is guaranteed that the system may easily allow for future growth without possible performance degradation.

User Training Capability

- This project must provide the capability for thorough end-user training, with three or more training sessions held within one month after the deployment date.

State: Mandatory

- This would ease user adoption and also ensure that users become proficient in the use of the new system efficiently.

Security Capability

- The system shall implement security features, to at least industry standards, including encryption and access controls, no later than three months after project close.

Implementation: Required

- This ensures protection of sensitive operational data and meeting regulatory requirements.

Predictive Analytics Capability

- The system shall deploy predictive analytics models to predict maintenance need and delays for the next six months. It shall be deployed in six months.

Implementation: Optional

- This will bring much value to the operational efficiency by enabling proactive management of maintenance and scheduling.

Cost Optimization Capability

- The solution shall include insights that allow a minimum operational cost reduction of 15% within the first year.

Implementation: Mandatory

- It focuses on financial efficiency through its reliance on facts and data analysis to guide decisions.

Change Management Capability

- The project should establish a change management process, which defines and documents all changes, with reviews of these bi-weekly during the life of the project.

Implementation: Mandatory

- This ensures that the changes are well managed and that all concerned stakeholders are on the same page regarding any change.

3.2 Operations, Administration, Maintenance and Provisioning (OAM&P)

The following are some common OAM&P capabilities for an aviation operational data system.

User Data Backup: Allow the user to schedule periodic automated backups of all operational data on ADLS to ensure data integrity and availability.

Fault Recovery: Allow for failover and disaster recovery in case of faults. Use builtin HA/DR services available in Azure for minimal downtown.
Routine Maintenance: Window maintenance schedules to update the system by optimizing and patching. It automates these updates through CI/CD pipelines in Azure DevOps. Monitoring and Alerting: Allows for fulltime monitoring with Azure Monitor and Azure Application Insights; provides alerts if something out of the ordinary happens, such as delays in data ingestion or failures on a data pipeline.
User Management: Azure Active Directory provides role based access to the resources so that the users can only access views and functions needed to perform their tasks.

3.3 Security and Fraud Prevention

To address internal and external security related issues:

Data Encryption: Data in transit and at rest will be encrypted by applying Azure's standard encryptions of TLS for data in transit and AES256 for data at rest.
Access Control: Utilize MFA and RBAC for an additional layer of security in accessing the system. Threat Detection: Leverage Azure Security Center to identify and mitigate the threat that may be presented in unauthorized access or DDoS attacks. Audit Logs: Log all the activities done on the system with minute detail using Azure Monitor and Log Analytics to track fraudulent activities in the system and prevent them.
Regular Security Audits: Perform security audits from time to time to detect and rectify vulnerabilities.

3.4 Release and Transition Plan

The release and transition plan will cover the following:

Pilot Deployment: Initial deployment in a controlled environment with selected users to validate the performance of the system.
Gradual Rollout: Phased deployment strategy to minimize disruptions. First, deploy to noncritical operations, then scale up.
Training and Documentation: Extensive users and administrator training; documentation of system operations and maintenance.

Feedback and Iteration: User feedback from the pilot and early rollout; assessment and fixing of issues before full deployment.
Post Implementation Support: A support team available to users for questions and issues that come up during the transition.

4 Project Design Description

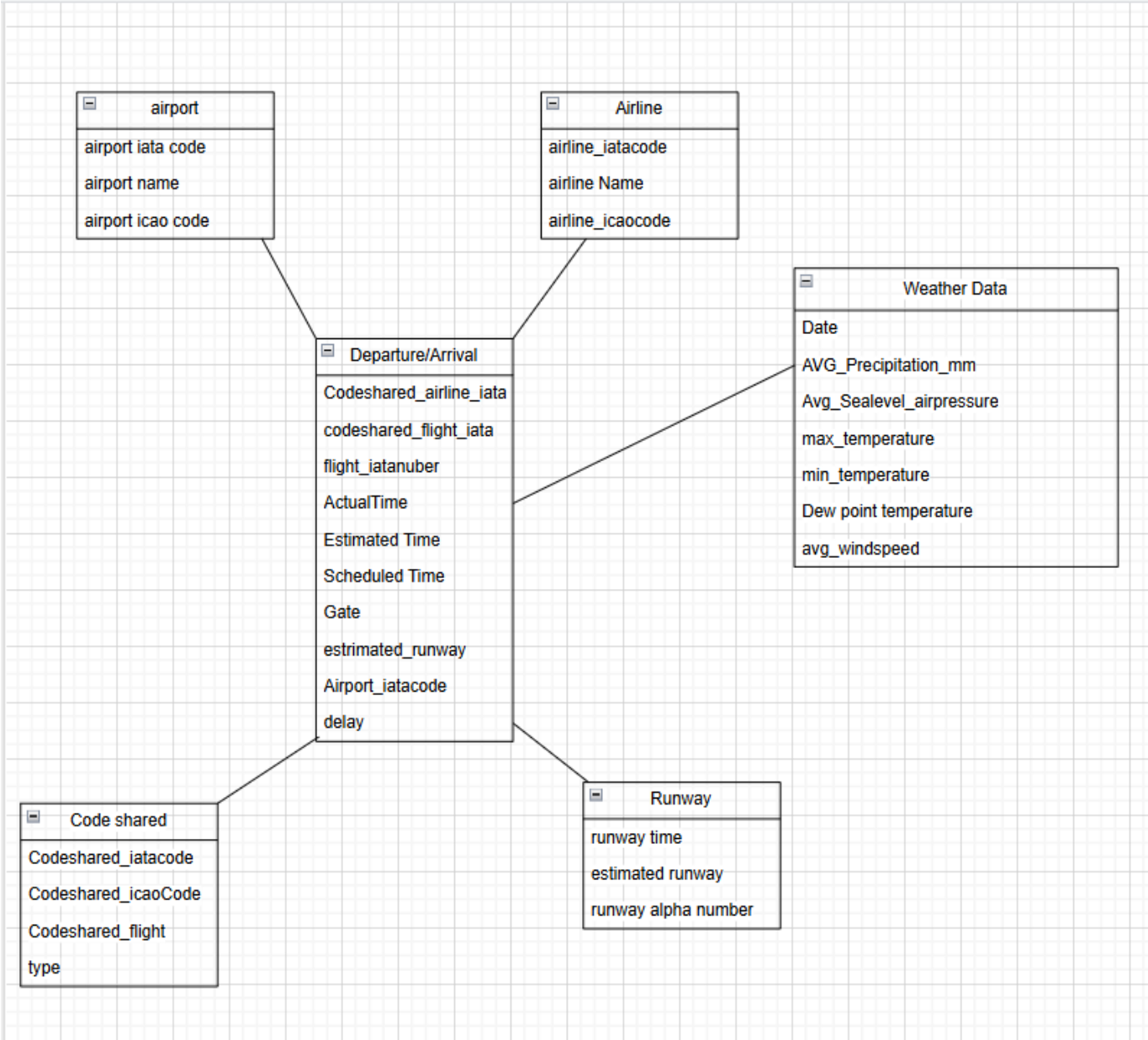
It demonstrates the complete integration of Azure services to design an efficient, scalable, and secure data processing system.

Components:

1. Azure Data Factory (ADF): This service was used for orchestrating ETL processes that pull data from APIs like flight scheduling, maintenance logs, and weather systems.
2. Azure Data Lake Storage (ADLS): This will provide a single area of storage for all operational data. It supports both structured and unstructured data.
3. Azure Databricks: This layer is responsible for data processing and its transformation into a star schema format that can be analyzed with ease.
4. Power BI: Visualizes data through dashboards and provides realtime insight into key performance metrics.
5. Azure DevOps: Maintains the CI/CD pipelines for the continuous integrations and deployments.

Data Flow:

1. Data Ingestion: ADF extracts data from a wide range of sources.
2. Data Storage: The data is to be stored in ADLS for long term purposes.
3. Data Processing: Databricks processes the raw data and structures it into star schema.
4. Data Presentation: Data, after transformation, will be presented on the Power BI dashboards for realtime monitoring and decision making.



5 *Internal/external Interface Impacts and Specification*

The project introduces several internal interface changes to seamlessly integrate data from a diverse set of sources. On the inside, data integration involves connecting the dot to the flight scheduling system for fetching current and historical schedules, interfacing aircraft maintenance logs and sensor systems for fetching operational and diagnostic data, and integrating with fuel management systems for tracking and optimization of fuel consumption. Secondly, the crew scheduling systems will be integrated to capture workforce utilization data for better resource planning. Processing will be done using Azure Data Factory for orchestration of ETL pipelines and Azure Databricks for complex data transformation. The unified data pipeline ensures compatibility with the star schema structure for optimized analytics performance. It will act as an Azure Data Lake Storage repository, requiring reevaluation in the ways of structuring and accessing both structured and unstructured data.

Externally, the project has major changes to third-party data interfaces. For example, the integration with external weather data APIs will allow real-time analytics on the impact of weather on flight operations. Interfaces with regulatory systems ensure that standards such as aviation industry standards on crew duty hours and maintenance records are complied with. Updates to user-facing applications include enhancements to the dashboards and reports that will be used by clients for improved operational insights and predictive analytics. Furthermore, it will develop secure APIs to enable aviation businesses to pull out subsets of data for decision-making and reporting purposes internally. This will help smoothly transition to the new system.

Similarly, the shared data structures will see transformative updates. It will be updated with an integrated star schema. This will collapse the different models into one with no redundancies but consistency across the board. Dimension and fact tables will be designed to support comprehensive analytics on metrics related to flight delays, fuel efficiency, and crew utilization. Data access protocols will be enhanced to real-time query facilitation, with security measures strengthened for controlled access to the unified data repository. These changes affect existing systems, which use either siloed or legacy structures, by demanding alignment with the new architecture.

One of the cornerstones of the system will be persistent data storage that will enable real-time and historical data to be retained for analytics and decision-making. The structured data would be stored in well-defined fact and dimension tables, logically categorized into routes, aircraft types, weather conditions, and time periods. Unstructured data, like sensor logs and maintenance records, will be kept in raw format for detailed analysis. The weather data will also be persisted from external APIs for the support of predictive models and analytics of long-term trends. Metadata and audit logs of changes will ensure traceability of data and compliance with governance standards.

These changes involve updates to user documentation, EISs, and Application Interface documents. The documentation needs to outline new data pipelines, APIs, and structures of storage so that everyone is on the same page concerning system design and functionality. This holistic approach ensures that the aviation operational data system delivers actionable insights and enhances efficiency across industries.

6 Design Units Impacts

The project's design impacts several functional areas, with each represented as a separate design unit. These are collectively specifying the new system architecture to meet the requirements of the project. The following is a detailed description of each functional area, its respective role, the impacts, and requirements.

6.1 Functional Area: Data Integration Unit

6.1.1 Functional Overview

The Data Integration Unit is responsible for consolidating data from a wide variety of internal and external systems. This ranges from API connections to the retrieval of raw data via sources such as flight scheduling systems, maintenance logs, weather APIs, fuel management systems, and crew scheduling databases. In this process, Azure Data Factory is used to orchestrate ingestion in order to ensure data validation and normalization before storage for further processing and analysis.

6.1.2 Impacts

The system needs an integration of new interfaces to connect with external data sources, such as weather and regulatory systems, where existing internal interfaces need to be updated toward the centralized framework. This requires setting up API connections in such a way that the data security and governance standards are ensured.

6.1.3 Requirements

This unit covers the requirements necessary for integrating operational systems; establishing secure API connections; and ingesting, validating, and normalizing data to prepare it for downstream processes.

6.2 Functional Area: Data Storage and Management

6.2.1 Functional Overview

The Data Storage and Management Unit manages the centralized storage of both structured and unstructured data through Azure Data Lake Storage. This functional area provides scalability and durability to the system for large volumes of aviation data that are generated daily. Structured data will be organized in a star schema format for optimized analytics, while unstructured data, including sensor logs and maintenance records, will be stored raw for more in-depth analysis.

6.2.2 Impacts

This will affect the legacy data systems, which have to migrate or integrate with the new storage architecture. Hierarchical tagging and metadata management are to be implemented at this level to make the organization and retrieval of stored data efficient.

6.2.3 Requirements

Requirements include central storage for operational data, a star schema for structured data, and support for unstructured data storage.

7 Open Issues

As of the current draft, there are no open issues .The document has been reviewed and is considered completed for the current stage of development .Any future updates or changes will be addressed in subsequent versions of the document

8 Acknowledgements

The document was authored by **DINESH VALLERU** with the assistance of **VENKATA JASWANTH REDDY AKKAMREDDYGARI, VENKATA NAGA SAI HARSHIT MANIKONDA, KODANDA RAMU YANAMALACHINTALA.** Special thanks to **STEVE SHIH** for their valuable contributions and inputs throughout the project. Their expertise and feedback have significantly contributed to the clarity and comprehensiveness of the development

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10 Appendices

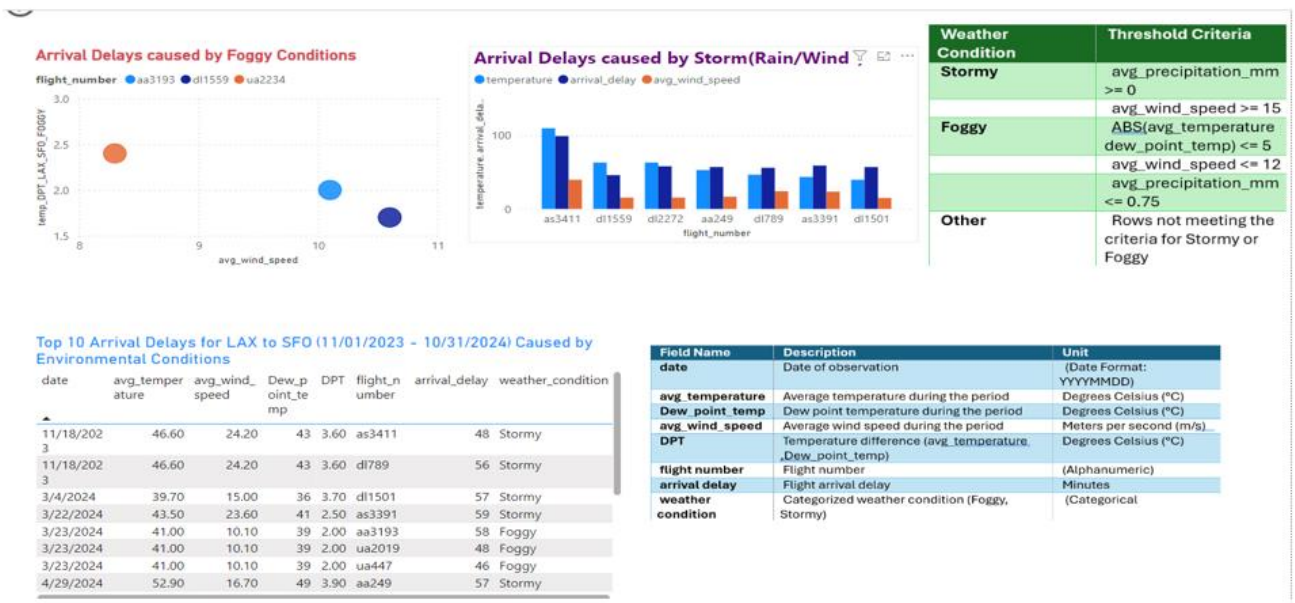
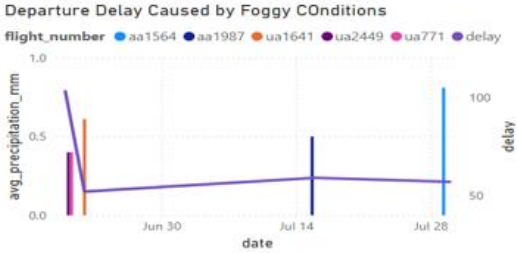


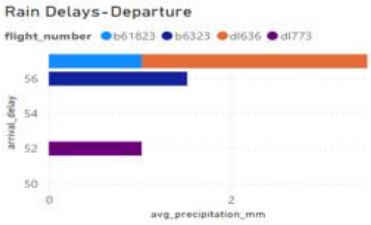
Figure 3-Arrival Delays LAX to SFO



Summarized Details

date	dew_point_temp	avg_wind_speed	avg_temperature	visibility	flight_number	arrival_delay	weather_condition
1/9/2024	23.00	13.70	24.60	1.50	aa2012	54	Rainy
4/6/2024	49.10	6.20	50.90	4.80	aa1597	54	Rainy
5/21/2024	63.00	7.50	64.80	6.00	aa1987	50	Rainy
5/21/2024	63.00	7.50	64.80	6.00	ua2095	56	Rainy
6/22/2024	71.10	11.20	74.10	3.20	ua1641	52	Foggy
6/20/2024	71.60	9.30	76.10	3.50	ua2449	53	Foggy
6/20/2024	71.60	9.30	76.10	3.50	ua771	50	Foggy
7/30/2024	72.50	8.70	78.30	4.00	aa1564	57	Foggy
7/16/2024	73.40	7.50	80.20	2.50	aa1987	59	Foggy

Figure 4-DFW-MIA



Weather Condition	Threshold Criteria
Foggy	Absolute difference between avg_temperature and dew_point_temp ≤ 4 - avg_wind_speed ≤ 14 avg_precipitation_mm ≤ 0.75
Snow	avg_temperature ≤ 10 OR min_temperature ≤ 5 avg_precipitation_mm ≥ 0.75 - avg_wind_speed ≥ 15
Rainy	avg_precipitation_mm ≥ 1.0 avg_temperature > 10

Top 10 Arrival Delays for JFK to LAX (11/01/2023 - 10/31/2024) Caused by Environmental Conditions

date	avg_precipitation_mm	avg_sealevel_airpressure	avg_temperature	avg_wind_speed	dew_point_temp	max_temperature	min_temperature	weather_condition
1/4/2024	0.25	1,025.90	-3.00	18.50	-5.00	1.00	-6.50	SNOW
1/20/2024	2.03	1,013.80	-2.50	22.00	-4.00	1.50	-6.00	SNOW
1/16/2024	1.52	1,031.50	-1.20	20.00	-3.00	2.00	-4.50	SNOW
11/18/2023	0.76	1,027.20	6.50	15.00	3.00	10.00	3.00	SNOW
11/11/2023	1.02	1,018.70	8.00	12.00	5.00	12.00	4.00	SNOW
9/11/2024	0.51	1,016.00	22.80	10.00	19.00	27.30	18.60	Foggy
6/16/2024	1.02	1,012.80	24.00	15.00	21.50	29.50	19.00	Rainy
6/11/2024	0.25	1,013.50	23.50	14.00	20.30	28.40	19.50	Foggy
7/29/2024	1.52	1,010.30	27.50	12.50	23.20	32.00	23.50	Rainy
8/1/2024	3.05	1,009.50	28.70	18.00	24.00	33.20	24.60	Rainy

Figure 5 JFK -LAX